

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

EX-8303 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
FACTS
(Elective - III)
Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain the term FACTS.
b) Define reactive power.
c) Differentiate between reactive power absorbers and suppliers.
d) Explain in detail about series and shunt compensation in transmission line.

OR

Explain in detail about the classification of different FACTS controller.

2. What are the factors which limit loading capability?
3. a) Describe the basic concepts of controlled series compensator.
b) Explain the operation of TCSC with the help of suitable diagram.
4. What is unified power flow controller (UPFC)? Explain the principle of operation characteristics along with control its performance.
5. a) What do you know about modeling of GCSC.
b) Describe the application of GCSC.
6. Explain the operation of SSSC and give the comparison between Variable Series Compensation and SSSC.
7. Describe the construction, principle of operation and V-I characteristics of STATCOM.
8. Write short notes on (any two):
 - a) Need of reactive power compensation in Power system
 - b) SSR mitigation
 - c) Need for facts controller
 - d) Prevention of voltage instability
